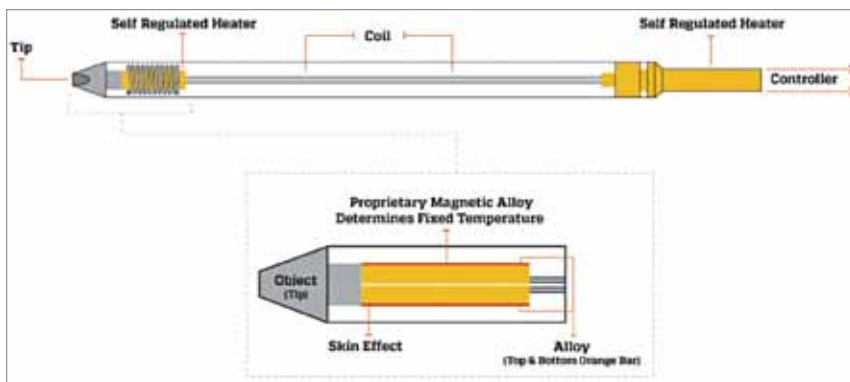




Metcal SmartHeat Technology



TMT R9900S Inline Robot Soldering System
(Credit: www.thermaltronics.com)

dering machines that feature linear motion in 3-axis and 4-axis type soldering robots are currently available in the market. In 3-axis, the soldering tip moves in 3 directions—X, Y and Z, with Z as the movement of the base plate. Similarly, in 4-axis type the tip moves in 3 directions—X, Y and Z, and in an angular momentum of the soldering tip. Here, the soldering tip can change its angle to match any angular surface on a PCB.

The 6-axis soldering works a bit differently from the 3-axis and 4-axis types. Conventionally, a solder wire is fed to the soldering tip to perform a solder joint on a PCB. However, as per the IPC soldering standards, the solder wire must be fed to the joint and not to the soldering tip. This type came up as a solution to

comply with IPC standards where the robot has two independent mechanisms—one to feed the soldering wire to the joint directly and the other to solder on to the PCB.

The latest upgrade is the 8-axis soldering that works slightly differently from the other types, giving 360 degrees of freedom. This increases the capabilities of the robot to perform operations in fewer setups. Considering that each setup requires additional power, material, and time, fewer setups translate into cost-savings. Thermaltronics Inc's TMT RS0099S is a model with 8-axis capability and 360° independent fixture rotation.

Laser soldering. Laser soldering is a non-contact process that eliminates the risk of marring or damaging sensitive components. This technique gives flexibility and is found to be suitable for mass production. As the system has a completely closed structure, it is free of pollution and ensures safety. It also provides process enhancements and zero consumable expenses.

The recent launches have various techniques that use different laser beam shapes like ring

type or square type. It gives flexibility for manufacturers to choose the shape considering their board pattern and components to avoid burning sensitive parts. A few machines have also come up with beam-splitting technology providing twin-spot lasers to simultaneously solder two joints. Japan Unix Laser Soldering Equipment is among the recent launches featuring various patented laser beam shapes suitable for micro-soldering applications.

Micro-soldering. Some components used in a PCB board are smaller than the size of a grain and micro-soldering uses smaller tools to solder them. One can connect even small wires that are nine microns in size. Laser soldering systems are found to be effective for micro-soldering.

"A laser's parameter is much smaller than even the finest soldering tip. It can reach spaces that are too narrow or inaccessible for an iron tip. As electronic parts become smaller and more intricate, this is becoming more relevant," says Shailendra Mathur, business head at Maxim SMT Technologies.

Advanced software. CAD/CAM software has become increasingly democratised, more affordable, and accessible. The recent software systems allow robots to be set up with minimal programming in a few steps.

These software can be easily integrated with real-time analysis, model based definition (MBD) software, and 3D viewing systems, thereby enabling quality improvements with reduced chances of user errors.

Apollo Seiko's L-CAT NEO-N is one of the soldering robots that allows reduction in cycle time and makes programming and setup much simpler. Robot teaching can be performed by the touch panel teaching pendant, a computer, or a tablet.

Dual port operating



Japan Unix Soldering Equipment